

Pressure Derivatives of Elastic Constants of Single Crystal MgO and MgAl₂O₄

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Ultrasonic velocities of single crystal MgO (periclase) and MgAl₂O₄ (spinel) were measured up to 7.78 GPa and 6.16 GPa, respectively, under hydrostatic pressure. The present success was attained by making various technical developments, such as a liquid-solid hybrid system and indium bond between sample and buffer-rod. Analytical procedures were also developed both in required corrections for apparent travel times, and in formulation of pressure effect on elasticity. A distinct shear softening was found in MgAl₂O₄, and was predicted to occur in MgO beyond 30 GPa. Isothermal equation of state of MgO was derived by combining the present results with shock hugoniot. Heat capacity, Grüneisen constant, and coefficient of thermal expansivity of MgO at high pressures were calculated by means of a new model equation of state. Negative thermal expansivity is expected to take place in MgO above ~50 GPa. The pressure derivative of rigidity of γ -Mg₂SiO₄ is estimated to be 0.56–1.0 by means of a lattice dynamic analogy with MgAl₂O₄; the lower margin of the estimate leads to the pressure-induced shear softening of γ -Mg₂SiO₄ in the earth's upper mantle.

1. Introduction

Elastic constants and their pressure derivatives of the mantle materials are very important to predict the effect of pressure on crystal lattice of minerals, and to construct mineralogical models of the earth's interiors consistent with seismological information. However, such experimental data have thus far been unsatisfactory in respect to accuracy because of the limited pressure range of experiment.

Kozuki *et al.* (1986) developed a liquid-solid hybrid system to generate hydrostatic pressure up to ~8 GPa, which is twice the maximum hydrostatic pressure (~4 GPa) attained in piston cylinder apparatus (e.g., Fujisawa *et al.*, 1981; Kondo *et al.*, 1981; Jackson and Niesler, 1982). The extended pressure capability allows more accurate investigation on the pressure dependence of elastic constants than those done hitherto. By use of the liquid-solid hybrid system, I studied the pressure dependence of elasticity on two important minerals, MgO (periclase) and MgAl₂O₄ (magnesium aluminate spinel).

Because MgO is a representative oxide mineral, as well as a candidate mineral constituting the lower mantle, the elasticity of MgO has been extensively studied by

Received October 18, 1989; Accepted May 10, 1990

various methods, such as high-pressure ultrasonic experiments (Bogardus, 1965; Anderson and Andreatch, 1966; Chang and Barsch, 1969; Spetzler, 1970; Ito *et al.*, 1977; Bonczar and Graham, 1982; Jackson and Niesler, 1982), shock-wave compressions (Al'tshuler *et al.*, 1965; McQueen and Marsh, 1966; Vassiliou and Ahrens, 1981), and static compression in diamond anvil cell (DAC; Mao and Bell, 1979). Only the ultrasonic experiments have provided information on the pressure dependence of elastic constant. However, a discrepancy larger than 10 % is obvious among the reported values for the first pressure derivatives (Table 4). It is evident that uncertainty of 10% causes serious ambiguity when the elasticity of MgO is extrapolated to the lower mantle conditions. Therefore more-accurate data are required for the pressure dependence of elasticity of MgO.

Magnesium aluminate spinel MgAl_2O_4 is a member of A_2BX_4 compounds with the spinel structure Fd3m (e.g., Tokonami and Horiuchi, 1980). The elasticity of spinel-structured minerals has received special attention, because the modified spinel (β) and the spinel phases (γ) with the composition of $(\text{Mg,Fe})_2\text{SiO}_4$ are major constituents of the mantle transition zone with depth of 400–650 km (e.g., Ringwood and Major, 1970; Akimoto *et al.*, 1976). Weidner *et al.* (1984) determined a complete set of elastic constants of $\gamma\text{-Mg}_2\text{SiO}_4$ by means of Brillouin scattering method under atmospheric pressure with a micro-crystal (~ 0.1 mm in edge length) synthesized by Sawamoto (1986). The pressure dependence is still unknown, because a single crystal of $\gamma\text{-Mg}_2\text{SiO}_4$ with a sufficient size for ultrasonic measurement has not been available. However, the pressure effect on elasticity of MgAl_2O_4 gives some significant insight into that of $\gamma\text{-Mg}_2\text{SiO}_4$. The elasticity of MgAl_2O_4 under high pressure has been studied only once up to 1 GPa in gas confining medium by Chang and Barsch (1973) by means of ultrasonic method. Considering the limited pressure range in their study, the pressure effect on elasticity of MgAl_2O_4 should be re-examined up to higher pressures.

In the present study, ultrasonic measurement was carried out under hydrostatic environment up to 6–8 GPa in order to obtain accurate pressure dependence of the elastic constants of MgO and MgAl_2O_4 .

In this paper, I report the results with the experimental details and newly developed data-processing methods. Extrapolation of the elastic constants based on the present results suggest that shear softening might occur at higher pressures in both MgO and MgAl_2O_4 . The equation of state was examined for MgO; a negative thermal expansivity was predicted at pressures higher than 50 GPa. The geophysical importance of these features will be also pointed out.

2. Experimental

2.1 Hydrostatic pressure cell for ultrasonic study

The 8 GPa hydrostatic system has sample space of 150 mm^3 , which is sufficient for ultrasonic study with bulk sample of ~ 3 mm both in length and in diameter. The system is compressed in the MA 8 high pressure apparatus (e.g., Kawai and Endo, 1970; Yoneda *et al.*, 1984), which is driven by means of the RH3 guide blocks (Kumazawa *et al.*, 1972) in the 10 kt press installed at Nagoya University (Yoneda *et al.*, 1986). The homogeneous metal gaskets developed by Yoneda (1987) play an essential role in

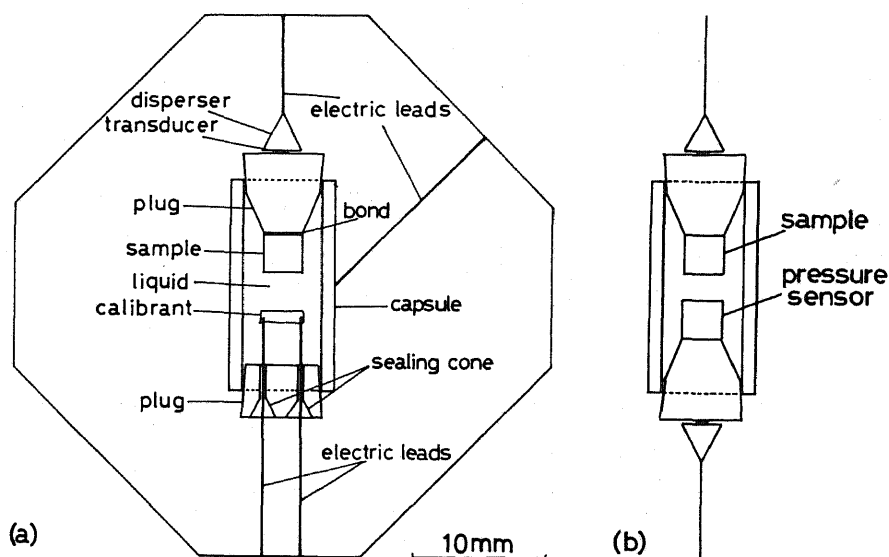


Fig. 1. (a) Cross section of a liquid-solid hybrid system before compression. A liquid cell is embedded in an octahedron made of semi-sintered magnesia with porosity of 19%, supplied by Mino Ceramic Co. Liquid medium is a mixture of methanol and ethanol (4 : 1 in volume ratio); capsule, stainless steel (6 mm, inner diameter; 8 mm, outer diameter); upper plug, hardened steel ($H_{Rc} \sim 60$); lower plug, stainless steel; disperser, brass. Indium is used as bond between sample and plug. The plug with transducer plays three roles: a plug itself for capsule, a buffer-rod for transmission of acoustic wave, and an electrode for transducer. The disperser disperses the backward wave from the transducer by means of its cone shape, and also serves as an electrode for transducer. Calibrant of bismuth is 0.5 mm thick. The sealing cones are filled with epoxy resin in order to prevent leakage of liquid (for detail, Kozuki *et al.*, 1986). (b) Cell assembly involving double acoustic system, where bismuth calibrant is replaced by another acoustic system for P wave in the [100] direction of MgO; i.e., MgO pressure sensor.

preventing the anvils from severe deformation, and in decreasing the probability of the blowout (Yoneda, 1987).

Figure 1 shows the present cell assembly to measure elastic velocity under high hydrostatic pressure. The transducer placed outside of the plug was the main alteration from the previous cell assembly (Kozuki *et al.*, 1986), in which the transducer in the capsule was frequently out of order due to short-circuit and/or break of electric wire. The plug in the present assembly plays an important role of buffer-rod in acoustic system.

Hardened steel and tungsten carbide were tested for a plug material in the early stage of the present work. Although a tungsten carbide plug hardly deformed during compression in comparison with a plug of hardened steel, its high acoustic impedance was found to be disadvantageous to transmit wave energy to the present sample. The

hardened steel plug was used throughout the present study.

The pressure in the liquid cell was monitored by bismuth calibrant (Fig. 1a), or by MgO pressure sensor (Fig. 1b). Absolute pressure scale and calibration of pressure were of much concern in the present work, because they affect directly the resultant pressure derivatives of elastic constants. In the assembly with bismuth calibrant, the pressure was determined with transition pressure of bismuth. However, there still remains some uncertainty of the transition pressures (e.g., Decker *et al.*, 1972). Nomura *et al.* (1982) determined the transition pressures of bismuth (2.58 GPa for Bi I-II transition; 2.74 GPa for Bi II-III; 7.78 GPa for Bi IV-V) according to the NaCl scale (Decker, 1971), where both bismuth sample and NaCl powder were enclosed in a liquid cell. Therefore their data are the most reliable, free from ambiguity due to non-hydrostatic stress. On the other hand, in the assembly with MgO pressure sensor, pressure values were determined from the travel time of longitudinal wave of MgO along the [100] direction. The MgO pressure sensor was calibrated by bismuth calibrant twice; the reproducibility of ~ 1 ns in travel time corresponds to ~ 0.1 GPa in pressure. The pressure values based on the travel times were refined in the self-consistent iteration of MgO data (Sec. 4.1).

Figure 2 shows photographs of a liquid capsule. Although the capsule was squeezed considerably during compression, the final inner diameter was large enough to accommodate the sample. The recovered sample was found to be free from plastic deformation; the change in sample length was below detectable limit ($< 1 \mu\text{m}$). On the other hand, the recovered plug elongated by ~ 1 mm; nevertheless, it was hardly bent. In the pulse echo method adopted in the present study, such elongation of a plug does not affect the measured travel time at all.

2.2 Sample and transducer preparation

Single crystal MgO (~ 10 mm cube with [100] cleavage face) was grown by Tateho Chemical Co. Purity of the crystal was confirmed to be stoichiometric by electron probe micro-analysis. Powder X-ray diffraction gave a cell constant of 421.17 pm, or density of $3,584 \text{ kg/m}^3$. The crystal orientations were determined by Laue X-ray diffraction. A crystal was sliced into disks (~ 3.5 mm thickness) along the (100), (110) or (111) face. After polishing both surfaces of a disk, cylindrical specimens (~ 3 mm in diameter and ~ 3.5 mm in length) were cut out from the disk by means of ultrasonic machining.

Single crystal MgAl_2O_4 (~ 25 mm both in diameter and in length along [110] direction) was grown by Union Carbide Co. (USA). Stoichiometry and purity was confirmed by electron probe micro-analysis. Powder X-ray diffraction gave a cell constant of 808.36 pm, or density of $3,578 \text{ kg/m}^3$. Cylindrical specimens with [100] or [110] direction were prepared by the same procedures as those for MgO.

LiNbO_3 (lithium niobate) has been widely used as an electroacoustic transducer in ultrasonic measurements; Fujisawa and Ito (1984) demonstrated that it operates adequately even under non-hydrostatic pressure of 15 GPa. In the present work, Y-36° and Y-163° cuts were used for longitudinal and transverse modes, respectively (Warner *et al.*, 1967). After polishing a disk of LiNbO_3 (~ 1 mm in diameter and ~ 0.1 mm in thickness), it was stuck to a plug with epoxy resin.

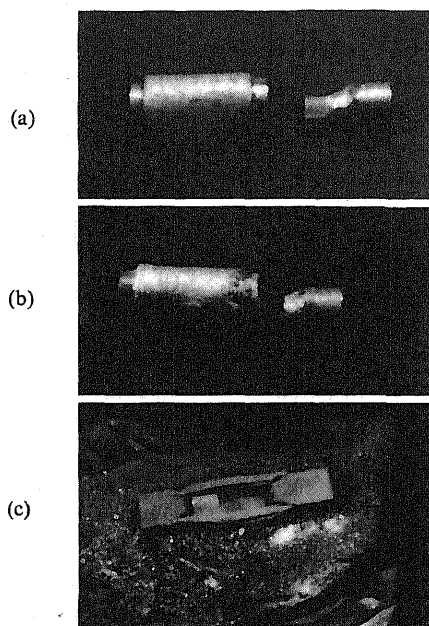


Fig. 2. Photographs of the sample cell. (a) Capsule and plug with sample before assembling the cell. (b) Recovered capsule after compression. (c) Cross section of the recovered cell involving a sample. The sample room is large enough to assemble all the parts in it.

2.3 Indium bond

Adhesive agents have been used as bond between buffer-rod and sample in order to get sufficient transmission of acoustic wave from buffer-rod to sample. Because properties of bond layer control the quality of observed signals, special care is required in selection and handling of bond material. In the present work, bond material must be (1) a good conductor of acoustic wave under high pressure, (2) of low shear strength under high pressure to protect sample from deviatoric stress, and (3) of high resistivity to corrosion by alcohols of the pressure medium. All the polymers tested failed in the above requirements.

Indium was found to be optimum as bond for the following reasons. (1) It has relatively high acoustic impedance ($18.5 \times 10^6 \text{ kg/m}^2\text{s}$ for longitudinal wave and $5.36 \times 10^6 \text{ kg/m}^2\text{s}$ for transverse wave at atmospheric pressure), which is advantageous to transmit wave energy to the present samples. (2) It has extremely low shear strength ($\sim 0.05 \text{ GPa}$ after King, 1965) even under high pressure, which is optimum to protect sample from deviatoric stress due to difference in compressibility among sample, buffer-rod, and bond. (3) It is not damaged by liquid medium of alcohols. (4) It has no phase transition in the present pressure range (e.g., Cannon, 1974). (5) It has a low melting point (156°C) and high affinity to minerals as solder.

Indium metal was coated on each face of the sample and the buffer-rod by means of a ultrasonic solder-iron (Sunbonder USM-III, supplied by Asahi Glass Co.) without any flux. Then, the sample and the buffer-rod were put together, pressed in a vice, and heated above melting point of indium. The final thickness of an indium layer was less than $\sim 5 \mu\text{m}$.

2.4 Measurement of acoustic velocity

Measurement of acoustic velocity was carried out by means of the pulse-echo-overlap (PEO) method (e.g., Papadakis, 1967). Figure 3 illustrates the timing chart and the block diagram of the PEO method. The carrier frequency was 40–60 Mhz for both longitudinal and transverse waves. The time resolution of ~ 0.5 ns in this system corresponds to $\sim 0.05\%$ of the travel time.

Multiple echo pulses are identified as shown in Fig. 4. T_{IL} is the difference of arrival time between the first echo pulse and the second one, and T_{LM} that between the second and the third. The intrinsic travel time is obtained from T_{IL} and T_{LM} by the correction method described in Sec.3.1 and Appendix A. Figure 5 shows examples of echo pulses on oscilloscope display.

It took ~ 10 min to determine a travel time, so that the measurement was made during a pause of the press drive in order to keep a constant pressure. Although the press load decreased slightly during the pause, the pressure in the cell did not decrease due to the frictional force among anvils and guide blocks (Yamamoto *et al.*, 1983). The measurements were commenced at the time 10 min after pump-off, because gradual decrease of travel time was observed in a few minutes just after pump-off. This variation

Fig. 3. (a) Timing chart of the pulse-echo-overlap (PEO) method. Oscilloscope is triggered by a pulse train with period of T , which should be equalized with the time difference τ between the two selected echoes. High-voltage pulse generator (HV pulser) is triggered by another pulse train divided from the above pulse train with ratio of 1/16–1/256. Luminosity of scanning line on oscilloscope display is modulated by a negative pulse with variable pulse width. The timing of the three pulses is adjusted by delays A and B. The two echoes used for measurement are selected to be seen on oscilloscope display by adjusting delay A, delay B, and pulse width of the luminosity modulator. (b) The typical examples of view of pulse echo overlapping. The spatial relation of the selected echoes (first and second in this case) on the display varies with the difference between T and τ . If $T = \tau$, the two selected echoes are found at the same position. (c) Block diagram of the instruments used in the present PEO method. The clock oscillator is a standard for controlling all the timing in the present measurement; one of the outputs of the clock oscillator is the oscilloscope trigger, of which the period is monitored by a counter. The matching circuit is introduced (1) to protect the instruments from HV pulse (~ 100 volt) and (2) to keep impedance matching for signal pulse (~ 10 microvolt) at the joint; the optimum matching is realized by adjusting variable resistors. A 50 ohm resistor near the transducer is necessary for impedance matching with 50 ohm co-axial cable.

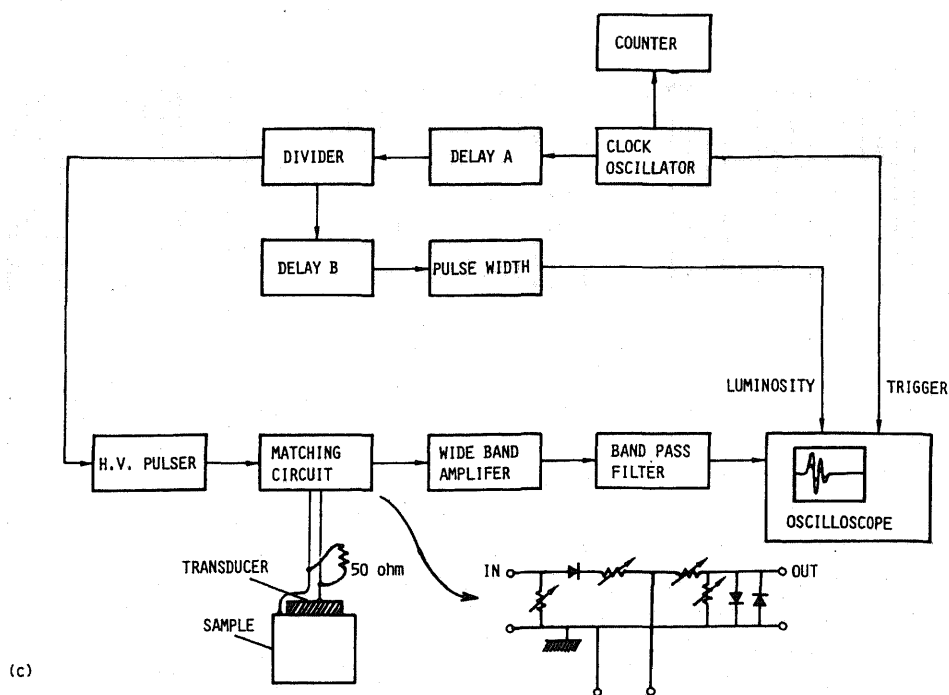
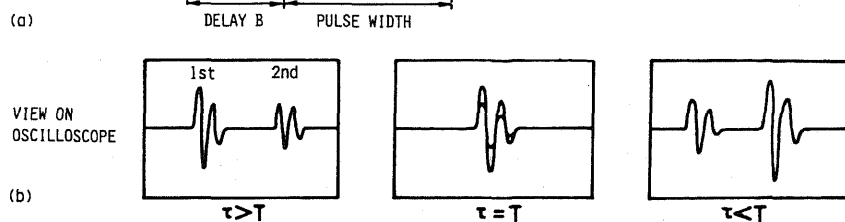
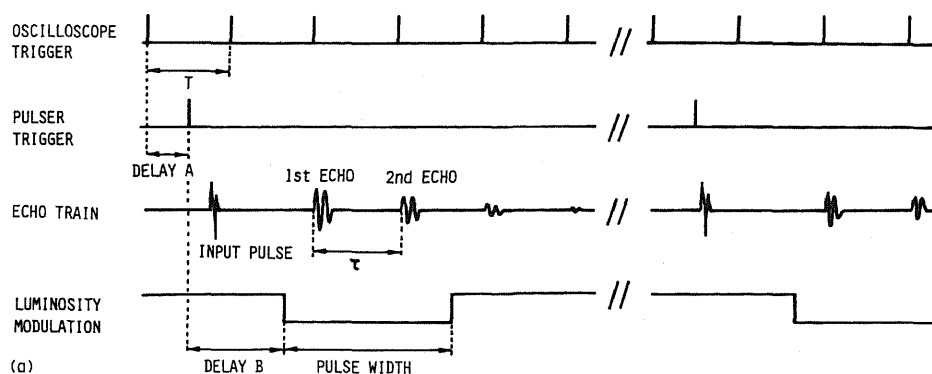


Fig. 3

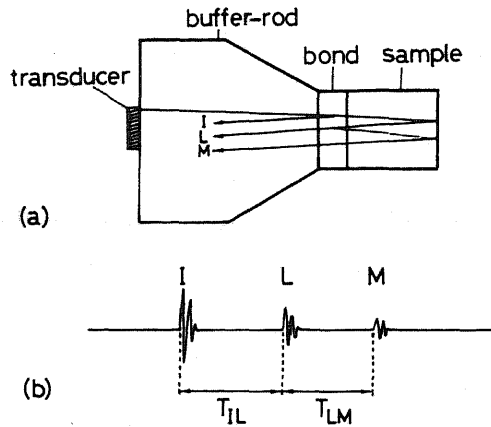


Fig. 4. (a) Ray trace of echo pulses I, L, and M in the present buffer-rod and sample. Thickness of bond is much exaggerated. Note that the multiple reflection at two ends of bond is not shown, for the sake of simplicity. (b) Schematic view of pulse train. Travel times of acoustic wave, T_{IL} and T_{LM} , are time delay of L to I and that of M to L, respectively.

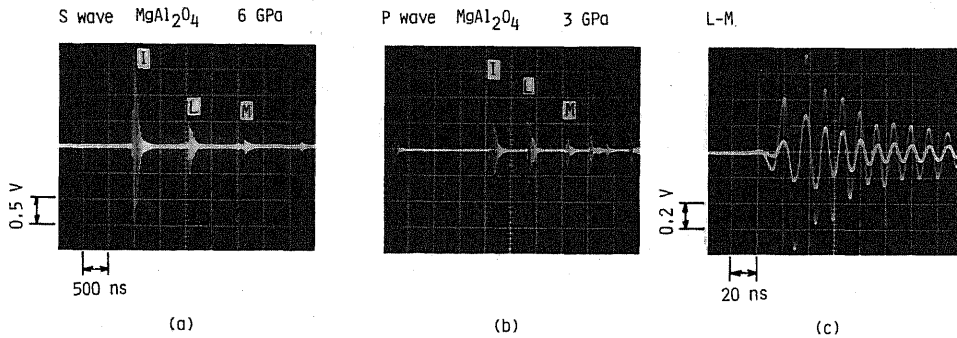


Fig. 5. Photographs of echo pulses. (a) Echoes of S wave. (b) Those of P wave. (c) Pulse-echo-overlapping of the echoes in (b).

of travel time may be due to adiabatic temperature rise of liquid and successive dissipation of it to the surrounding solid medium. The waiting time of 10 min was determined from experimental observations; it is much larger than the characteristic time (~ 5 min) of thermal relaxation in the pressure cell.

3. Data Analysis

3.1 Correction of measured travel time

It has been well known that the intrinsic travel time is affected by diffraction phase shift and bond phase shift (see Appendix A). Davies and O'Connell (1977) emphasized the necessity of eliminating these effects from the measured travel times in order to

Table 1. Ultrasonic velocities of MgO at atmospheric pressure. Uncorrected velocities in the present data correspond to the measured travel times of T_{LM} .

	[100]P	[111]P	[110]P	[100]S	[110]S ^a
Present data					
Uncorrected	9.136 (km/s)	10.173	9.936	6.606	5.348
Diffraction correction	9.127	10.171	9.928	6.600	5.340
Diffraction and bond correction	9.112	10.166	9.918	6.592	5.319
Previous data					
Bogardus (1965)	9.104	10.171	9.915	6.600	5.304
Anderson and Andreatch (1966)	9.099	10.165	9.909	6.597	5.303
Chang and Barsch (1969)	9.097	10.161	9.906	6.593	5.300
Spetzler (1970)	9.109	10.176	9.920	6.602	5.306
Sumino <i>et al.</i> (1976) ^b	9.129	10.193	9.937	6.604	5.311
Jackson and Niesler (1982)	9.100	10.163	9.908	6.593	5.302
Sumino <i>et al.</i> (1983) ^b	9.093	10.170	9.911	6.598	5.288

^a Velocity corresponding to C_s . ^b Acoustic velocities are calculated from the elastic constants obtained by means of a resonance method.

obtain precise pressure derivatives of elastic constants.

In the previous studies below 1 GPa, the diffraction effect has been suppressed by designing sufficient length of sample and/or buffer-rod (see Fig. A1). In the present study, however, the total length of sample and buffer-rod was only about 10 mm because of limitation in the sample volume of hydrostatic pressure cell. Therefore I estimated numerically the effect of phase shift due to diffraction from geometrical configuration (Appendix A.1).

On the other hand, the bond effect has been minimized usually by careful procedures both in bond preparation and in measurement of travel time. Further, Jackson *et al.* (1981) developed an offset method to correct the bond effect, where travel time was measured twice, in a one-transducer system and in a two-transducer system, at each datum point. Instead of these previous methods involving delicate and complicated procedures, I developed a self-consistent iterative method to eliminate the phase shift caused by the bond effect (Appendix A.2).

In Table 1, the uncorrected and corrected acoustic velocities of MgO at atmospheric pressure are compared with those obtained by previous workers. It is evident that the uncorrected values are slightly but systematically larger than most of the literature values, whereas the corrected ones are in good agreement. Therefore the correction for both diffraction and bond effects are essential in determining the pressure derivatives of elastic constants.

3.2 Analytical procedure

Selection of analytical formulas is important both in the analysis of experimental data and in the extrapolation of the results to higher pressures. The Murnaghan equation of state (EOS) and the Birch-Murnaghan EOS have been used widely as phenomenological EOS. Although the reliability of phenomenological EOS is still a subject of debate, the Murnaghan EOS has been used in analysis of pressure effect on elastic constants because of its simplicity. The Murnaghan EOS for an elastic constant C is

$$C = C_0 + C'P + \frac{1}{2}C''P^2, \quad (1)$$

where P is pressure, C_0 elastic constant at atmospheric pressure, and C' and C'' the first and the second pressure derivatives of the elastic constant at atmospheric pressure, respectively;

$$C' = \partial C / \partial P \big|_{P=0} \quad (2)$$

$$C'' = \partial^2 C / \partial P^2 \big|_{P=0}. \quad (3)$$

However, the author used the Birch-Murnaghan EOS in the present analysis, because it gives more reasonable extrapolation of elastic constants than the Murnaghan EOS does for every case examined (Yoneda, in preparation). In the Birch-Murnaghan EOS, the finite strain ε is introduced as a measure of volume contraction;

$$\varepsilon = (3/2)(V^{-2/3} - 1), \quad (4)$$

where V is volume normalized by V_0 (volume at atmospheric pressure). Note that the ε defined in Eq. (4) is 3 times that used by Birch (1952).

The following formulas (Eqs. (5)–(10)) are used in the analysis based on the Birch-Murnaghan EOS; the derivation of those equations is given in Appendix B (Eqs. (B.1)–(B.11)).

Pressure dependence of any elastic constant C is described explicitly in terms of ε ;

$$P = K_{T_0}(2\varepsilon/3 + 1)^{5/2}(\varepsilon + (\xi_{k_T}/2)\varepsilon^2 + (\eta_{k_T}/6)\varepsilon^3) \quad (5)$$

$$C = C_0(2\varepsilon/3 + 1)^{5/2}\{7\varepsilon/3 + 1 + (\xi_C/2)\varepsilon(9\varepsilon/3 + 2) + (\eta_C/6)\varepsilon^2(11\varepsilon/3 + 3)\}, \quad (6)$$

where K_{T_0} is isothermal bulk modulus at atmospheric pressure, and other parameters are written as

$$\xi_{k_T} = K'_T - 4 \quad (7)$$

$$\eta_{k_T} = K_{T_0}K''_T + K_T'^2 - 7K'_T + 143/9 \quad (8)$$

$$\xi_C = (K_{T_0}/C_0)C' - 4 \quad (9)$$

$$\eta_C = (K_{T_0}^2/C_0)C'' + (K_{T_0}/C_0)K'_TC' - 7(K_{T_0}/C_0)C' + 143/9. \quad (10)$$

Both MgO and MgAl₂O₄ belong to cubic symmetry with three independent elastic constants. In the present analysis, K_S (adiabatic bulk modulus), $C_S (= (C_{11} - C_{12})/2)$,

and C_{44} were chosen as the independent elastic constants. In the analysis, K_{S_0} , C_{S_0} , and $(C_{44})_0$ were fixed to the values obtained at atmospheric pressure in the present work, and six parameters (ξ_{K_S} , η_{K_S} , ξ_{C_S} , η_{C_S} , $\xi_{C_{44}}$, $\eta_{C_{44}}$) were determined by minimizing the sum of residual squared, S ;

$$S = \sum_i (C_{\text{obs}}^i - C_{\text{cal}}^i)^2, \quad (11)$$

where the suffix i denotes a datum point, C_{obs} and C_{cal} are observed and calculated stiffness, respectively.

The data of apparent travel time are not sufficient to calculate the elastic constants under high pressure, because the sample length changes with pressure. The length correction requires the conversion of K_S to K_T ; K_S is related to the measured travel time of acoustic wave, while K_T is related to the length contraction of sample. The conversion relation is

$$K_T = K_S(1 + \beta\gamma T)^{-1} = \kappa K_S, \quad (12)$$

where κ is a parameter introduced for simplicity; β , coefficient of volume thermal expansivity; T , absolute temperature; and γ , Grüneisen constant given by

$$\gamma = V\beta K_S / C_p \quad (13)$$

with specific volume V ($=1/\rho$; ρ : density) and heat capacity at constant pressure C_p . Pressure dependence of κ was calculated by use of Eqs. (B.15)–(B.18); the required physical constants are compiled in Table 2.

Several methods have been developed to correct sample length from the pressure dependence of travel time of acoustic wave (e.g., Cook, 1957; Thurston and Brugger,

Table 2. List of physical constants of MgO and MgAl₂O₄ used in the data analysis.

	MgO	MgAl ₂ O ₄
ρ (kg/m ³)	3,584.0 ^a	3,578.0 ^a
K_S (GPa)	162.7 ^a	197.9 ^a
$\partial K_S / \partial P$	4.13 ^a	5.66 ^a
$\partial K_S / \partial T$ (GPaK ⁻¹)	-0.015 ^b	-0.015 ^c
β (K ⁻¹)	3.1×10^{-5} ^d	1.9×10^{-5} ^e
$\partial \beta / \partial T$ (K ⁻²)	6.7×10^{-8} ^d	3.8×10^{-8} ^e
C_p (heat capacity) (JK ⁻¹ kg ⁻¹)	923 ^f	815 ^g
γ	1.52	1.29
κ	0.986	0.993
K_T (GPa)	160.4	196.5
$\partial \kappa / \partial T$ (K ⁻¹)	-7.6×10^{-5}	-3.9×10^{-5}
$\partial K_T / \partial T$ (GPaK ⁻¹)	-0.027	-0.023
$\partial \beta / \partial P$ (GPaK ⁻¹)	-1.1×10^{-6}	-0.59×10^{-6}
$\partial C_p / \partial P$ (JK ⁻¹ kg ⁻¹ GPa ⁻¹)	5.7×10^{-9}	3.2×10^{-9}
$\partial \kappa / \partial P$ (GPa ⁻¹)	7.9×10^{-4}	3.1×10^{-4}

^a Present data. ^b Sumino *et al.*, 1983. ^c Chang and Barsch, 1973. ^d Suzuki, 1975. ^e Suzuki *et al.*, 1979. ^f Watanabe, 1982. ^g Bonnickson, 1955.

1964; Spetzler *et al.*, 1972). In the present analysis, a self-consistent iteration was made in order to determine alternately a set of ξ_{k_T} and η_{k_T} and a set of finite strains ε for each experimental pressure; the first cycle of the iteration started at the calculation of ε for each data point with the values of $\xi_{k_T}=0$ and $\eta_{k_T}=0$; the length contraction of sample is given by $(2\varepsilon/3+1)^{-1/2}$.

4. Results and Discussion

4.1 Elastic properties of MgO

Table 3 summarizes the observed travel time for MgO after both the diffraction correction and the bond correction. The experimental data were analyzed in the two models. The first model involves both the first and the second parameters (ξ and η) in Eqs. (5) and (6). The resultant parameters are summarized as #1 in Table 4. The errors in η 's in #1 result may allow $\eta=0$ for all elastic constants. Therefore the second model involves only ξ (i.e., $\eta=0$) for all elastic constants. The results are indicated as #2 in Table 4. Significant difference is not found between #1 and #2 for values of the parameters. The calculated travel times, τ_{cal} , are compared with the observed ones, τ_{obs} , in Table 5, together with the pressure values. It is found that the residuals between τ_{obs} and τ_{cal} are within $\sim 0.35\%$ of τ_{obs} in both analyses of #1 and #2.

The values of C , C' , and C'' obtained in the present study are compared with those by previous workers in Table 4. All the elastic constants listed are in good agreement. The first pressure derivatives are also consistent with one another, considering the uncertainty contained in each determination. As to the second derivatives, the analyses #1 and #2 give somewhat different values. The present pressure derivatives are most reliable, because the measurement was carried out up to 7.78 GPa, which is far higher than the 1–3 GPa in the previous studies.

Figure 6 shows the extrapolation of elastic constants to higher pressures using parameters obtained in cases of #1 and #2. It is remarkable that both #1 and #2 predict softening of C_{44} in MgO above ~ 30 GPa. The geophysical importance of the pressure-induced shear softening will be pointed out in the last chapter.

The pressure-induced softening of C_{44} in the B1-structured material has been predicted by the lattice dynamic theory in connection with the B1-B2 transition (e.g.,

Table 3. Travel times of acoustic wave in MgO after the corrections. The measurement for [100]P mode was made twice. For [100]P and [100]S modes, pressures were determined by bismuth calibrant. For the other modes, pressures were monitored by MgO pressure sensor (3.636 mm in length); the travel times after the corrections are shown in parentheses under τ_{100P} .

[100]P (3.636 mm)		[110]P (3.526 mm)		[111]P (3.615 mm)		[100]S (3.636 mm)		[110]S (3.527 mm)	
P (GPa)	τ (ns)	τ_{100P}	τ (ns)	τ_{100P}	τ (ns)	P (GPa)	τ (ns)	τ_{100P}	τ (ns)
2.58	772.3 772.2	(776.4)	698.2	(779.2)	701.1	2.58	1,096.8	(777.1)	1,285.2
2.74	770.0 770.9	(750.7)	685.2	(762.8)	693.5	2.74	1,095.9	(749.6)	1,231.4
7.78	725.6 725.1	(736.1)	676.8	(737.3)	681.4	7.78	1,085.7	(736.9)	1,210.3

Table 4. Results for MgO in comparison with those by previous researchers.

	Mode					
	K_S	C_S	C_{44}	C_{11}	C_{12}	K_T
Present work						
C_0 (GPa)	162.7 ± 0.2	101.3 ± 0.13	155.8 ± 0.15	297.8	95.1	160.4
#1						
C'	4.13 ± 0.30	3.48 ± 0.18	1.31 ± 0.17	8.76	1.81	4.21
C'' (GPa $^{-1}$)	0.001 ± 0.10	-0.02 ± 0.07	-0.09 ± 0.05	-0.03	0.02	0.008
ξ_c	0.072 ± 0.29	1.50 ± 0.28	-2.7 ± 0.17	0.72	-0.94	0.21
η_c	4.7 ± 17	-4.8 ± 17	-2.6 ± 9	0.4	11.5	5.4
#2						
C'	4.24 ± 0.07	3.38 ± 0.03	1.29 ± 0.04	8.75	1.99	4.32
C'' (GPa $^{-1}$)	-0.029	-0.006	-0.075	-0.038	-0.026	-0.022
ξ_c	0.18 ± 0.07	1.35 ± 0.05	-2.68 ± 0.04	0.71	-0.65	0.32
Jackson and Niesler (1982)						
C_0	162.5	100.8	155.8	296.8	95.3	
C'	4.13	3.78	1.11	9.17	1.61	
C''^*	-0.058	-0.045	-0.032	-0.118	-0.028	
Spetzler (1970)						
C_0	162.8	100.9	156.2	297.4	95.6	
C'	3.85	3.64	1.09	8.70	1.42	
Chang and Barsch (1969)						
C_0	162.3	100.8	155.8	296.6	95.1	
C'	4.27	3.67	1.12	9.16	1.82	
Anderson and Andreatch (1966)						
C_0	162.3	100.8	156.0	296.7	95.1	
C'	4.49	3.75	1.16	9.48	1.99	

* The values are twice those in their paper, according to personal information.

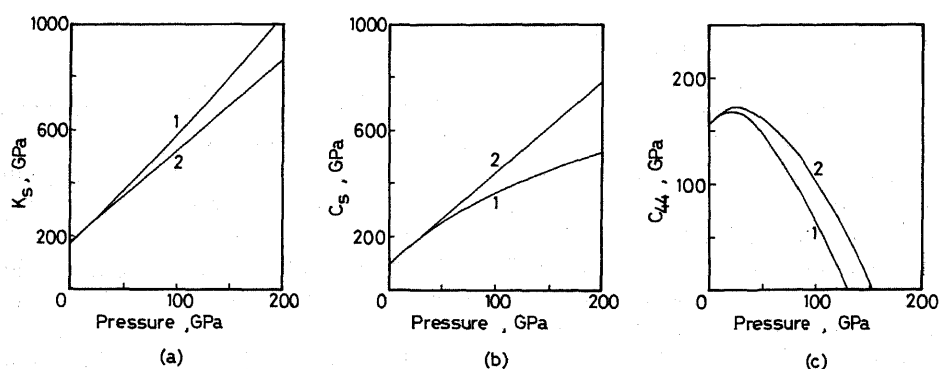


Fig. 6. Extrapolation of stiffness of MgO. Numerals 1 and 2 denote the extrapolations using the set of parameters of #1 and #2 in Table 4, respectively. (a) K_S versus pressure. (b) C_S versus pressure. (c) C_{44} versus pressure. C_{44} undergoes distinct softening for both cases.

Table 5. Comparison between the observed travel times, τ_{obs} , and the calculated travel times, τ_{cal} , for MgO. τ_{obs} is the measured travel time after the corrections. #1 and #2 correspond to results in Table 4. The values for [110]P, [111]P, and [110]S modes are based on MgO pressure sensor (see text). The pressures in [100]P and [100]S modes are results of bismuth calibrant.

[100]P				[100]S					
P (GPa)		τ_{cal}		τ_{obs}	P (GPa)		τ_{cal}		τ_{obs}
#1	#2	#1	#2		#1	#2	#1	#2	
2.58		771.2	771.3	772.3	2.58		1,094.9	1,094.9	1,096.8
2.74		769.7	769.8	770.5	2.74		1,094.5	1,094.5	1,095.9
7.78		725.8	726.0	725.4	7.78		1,084.5	1,083.6	1,085.7

[110]P					[111]P				
P (GPa)		τ_{cal}		τ_{obs}	P (GPa)		τ_{cal}		τ_{obs}
#1	#2	#1	#2		#1	#2	#1	#2	
2.06	2.07	699.6	699.5	698.2	1.77	1.78	702.8	702.7	701.1
4.80	4.82	685.5	685.2	685.2	3.46	3.47	695.4	695.1	693.5
6.50	6.53	677.4	677.1	676.8	6.35	6.38	683.8	683.4	681.4

[110]S				
P (GPa)		τ_{cal}		τ_{obs}
#1	#2	#1	#2	
1.98	1.99	1,286.5	1,287.3	1,285.2
4.92	4.93	1,234.4	1,235.5	1,231.4
6.40	6.43	1,210.8	1,211.6	1,210.3

* Average of the results of the two runs shown in Table 3.

Anderson and Liebermann, 1970; Okai, 1985). In Table 6, the first pressure derivatives of the B1-structured alkali earth oxides are listed with pressure values for the B1-B2 transformation. The B1-B2 transition in CaO was observed at ~ 60 GPa with volume reduction of 10 % by means of shock hugoniot and static compression with X-ray measurement (e.g., Jeanloz *et al.*, 1979), and that of SrO was detected at 36 GPa with volume reduction of 11% by Sato and Jeanloz (1981) in DAC. On the other hand, BaO undergoes two transitions at pressures of 9 and 14 GPa to a tetragonal structure with total volume reduction of 13 % (Liu and Basset, 1972).

The MgO is expected to undergo the transformation to the B2-structure at higher pressure than CaO and SrO, because Mg^{+2} ion has a smaller radius than Ca^{+2} and Sr^{+2} ions. However, the pressure-induced transition in MgO was not observed by X-ray study in DAC to 90 GPa (Mao and Bell, 1979). Shock hugoniot of MgO has been studied to ~ 260 GPa (Al'tshuler *et al.*, 1965; McQueen and Marsh, 1966; Vassiliou

Table 6. Summary of the elasticity, ion radius, and the B1-B2 phase transition in alkali earth oxides. Elasticity data of MgO are the present results. Those of CaO, SrO, and BaO are after Chang and Graham (1977). Those of MgO by Chang and Barsch (1969) are also shown in parentheses.

	MgO		CaO	SrO	BaO
$\partial K_s/\partial P$	4.13	(4.27)	4.83	5.18	5.52
$\partial C_s/\partial P$	3.48	(3.67)	4.14	4.33	3.50
$\partial C_{44}/\partial P$	1.31	(1.12)	0.196	-0.206	-0.472
Ion radius of cation					
in six-fold coordination (pm)	72	100	113	136	
Transition pressure (GPa)	<130*	60	36	9 (14)**	
Volume reduction (%)	—	10	11	5 (7)**	

* Predicted from shear softening of C_{44} along #1 in Fig. 6. ** BaO changes to a tetragonal phase through two-stage transition. The transition pressure and the volume reduction for the second transition are shown in the parentheses.

and Ahrens, 1981). Although Vassiliou and Ahrens (1981) concluded that no anomaly is found in their shock hugoniot up to ~ 200 GPa, they pointed out that a high density reported by Al'tshuler *et al.* at ~ 260 GPa may be caused by the B1-B2 transition.

Born's stability condition of cubic lattice requires simultaneously

$$K > 0, \quad C_s > 0, \quad \text{and} \quad C_{44} > 0. \quad (14)$$

When either C_s or C_{44} vanishes, the lattice becomes unstable and must change into another structure. This idea provides a useful criterion to estimate upper limit of transition pressure. The model EOS #1 suggests the transition pressure below 130 GPa; and the model EOS #2, that below 150 GPa (Fig. 6). If the extrapolations of C_{44} are reliable, it is quite probable that Al'tshuler *et al.* attained the B1-B2 transition in their shock experiment. The discrepancy of the transition pressure between ~ 260 GPa in shock experiment and ~ 150 GPa in the present prediction is not easily solved. However, possible explanations may be (1) the shock effect which usually gives higher transition pressure than that observed in static compression, and (2) the positive slope of the phase boundary in pressure-temperature space.

4.2 Equation of state of MgO

Because MgO is the most important oxide in the earth's mantle, I constructed the equation of state (EOS) of MgO by the following two methods: (1) The extrapolation of pressure-volume relation by substituting the parameters obtained in both models of #1 and #2 into the Birch-Murnaghan EOS; (2) The temperature correction of the shock hugoniot by using thermodynamic parameters calculated from the present results of elasticity (the resultant isothermal EOS is abbreviated as "shock-EOS").

In order to analyze shock hugoniot, it is necessary to evaluate heat capacity, Grüneisen constant, and thermal expansivity of the material. These quantities have been calculated by means of the Debye model so far. Recently, Kieffer (1979 a, b, c, 1980, 1982) proposed a new model superior to the Debye model for the cases of oxides

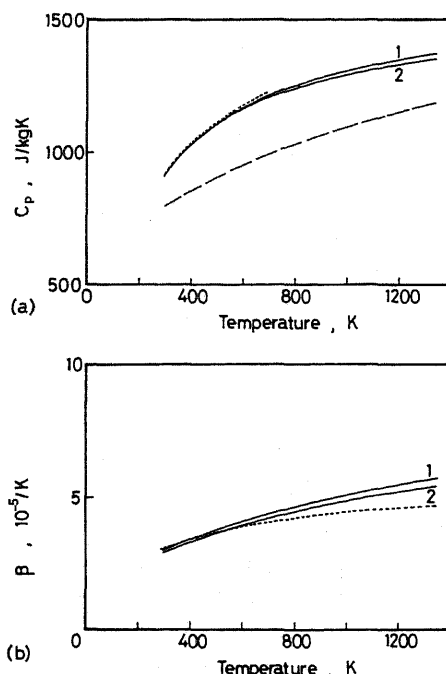


Fig. 7. (a) Temperature dependence of heat capacity at constant pressure C_p of MgO at atmospheric pressure. The C_p is given by C_v/κ ($\kappa = K_T/K_S$ in Eq. (12)), where the C_v is calculated by use of Eqs. (C.1)–(C.7). Solid lines are results of calculation made for the two cases of elastic parameters, #1 and #2, in Table 4. Experimental data (dotted line) is after Watanabe (1982). The results by means of the Debye model (dashed line) are shown for comparison. (b) Temperature dependence of coefficient of volume thermal expansivity, β , for MgO at atmospheric pressure. Calculation (solid line) is made for the two parameter sets, #1 and #2, by use of Eqs. (C.12)–(C.14). Experimental data (dotted line) is after Suzuki (1975).

and silicates, so the thermodynamic properties were calculated according to the Kieffer model. However, the original Kieffer model was found to be reasonably improved by introducing some simplification specific to the B1-structured material (see Appendix C for details).

The heat capacity at constant volume C_v , the Grüneisen constant γ , and the volume thermal expansivity β for MgO were calculated by use of Eqs. (C.1)–(C.7) and (C.12)–(C.14). Figure 7 shows the calculated heat capacity at constant pressure C_p and the β under atmospheric pressure. The calculated C_v , γ , and β under various volume-temperature conditions are exemplified in Table 7.

The linear relation between shock wave velocity U_s and particle velocity U_p was assumed;

Table 7. Example of calculated physical properties of MgO in shock-EOSs.

	P (GPa)	C_v (J/kgK)	$\beta(10^{-5} \text{ K}^{-1})$	γ
$v=1.0, T=300$ (K)				
#1	0.01	894.9	2.93	1.47
#2	0.01	894.9	2.88	1.44
$v=1.0, T=1,500$ (K)				
#1	7.30	1,219.7	4.32	1.47
#2	7.19	1,219.7	4.24	1.44
$v=0.85, T=300$ (K)				
#1	37.89	794.0	0.50	0.48
#2	37.71	787.2	0.81	0.75
$v=0.85, T=1,500$ (K)				
#1	41.13	1,211.4	0.95	0.59
#2	42.29	1,211.0	1.38	0.82
$v=0.75, T=300$ (K)				
#1	99.73	778.4	-1.85	-2.70
#2	93.27	741.1	-0.44	-0.59
$v=0.75, T=1,500$ (K)				
#1	87.17	1,208.1	-2.01	-1.82
#2	91.46	1,205.8	-0.28	-0.23

C_v is heat capacity at constant volume; β , coefficient of volume thermal expansivity; and γ , Grüneisen constant. #1 and #2 correspond to the results shown in Table 4.

$$U_s = C_b + S U_p, \quad (15)$$

where C_b ($=6,736.8 \text{ m/s}$) was bulk sound velocity ($(K_{s0}/\rho)^{1/2}$) of the present elasticity data, and S was determined to 1.34 ± 0.02 by means of a least square fit to the shock data (McQueen and Marsh, 1966; Vassiliou and Ahrens, 1981). The procedure of the temperature correction of shock hugoniot was the same as that in the previous works (e.g., Rice *et al.*, 1958; Jamieson *et al.*, 1982) except the method estimating C_v and the use of the generalized Mie-Grüneisen EOS, Eq. (C.11), instead of the Mie-Grüneisen EOS.

Figure 8 shows the various EOSs based on the present elastic data of MgO, together with those in previous studies. We find the following aspects for the #1 Birch-Murnaghan EOS: (1) It is consistent with the #2 Birch-Murnaghan EOS up to 40 GPa within difference of 1 GPa; (2) It is consistent with both the #1 and #2 shock-EOSs up to ~ 60 GPa within 1 GPa; (3) It shows better agreement with the results in DAC than the #2 Birch-Murnaghan EOS does. We conclude that the #1 Birch-Murnaghan EOS is superior to that of #2, and that it is reliable up to 60 GPa within accuracy of 1 GPa. On the other hand, the present shock-EOSs are more consistent with the results of direct compression in DAC than the previous shock-EOS by Jamieson *et al.*, which was rather compressible. The pressure-volume-temperature relation in the #1 shock-EOS is numerically listed in Table 8.

As shown in Tables 7 and 8, negative thermal expansion is expected for MgO above ~ 50 GPa. The negative thermal expansivity is associated with negative mode Grüneisen parameter of C_{44} mode. The similar anomaly is found in the

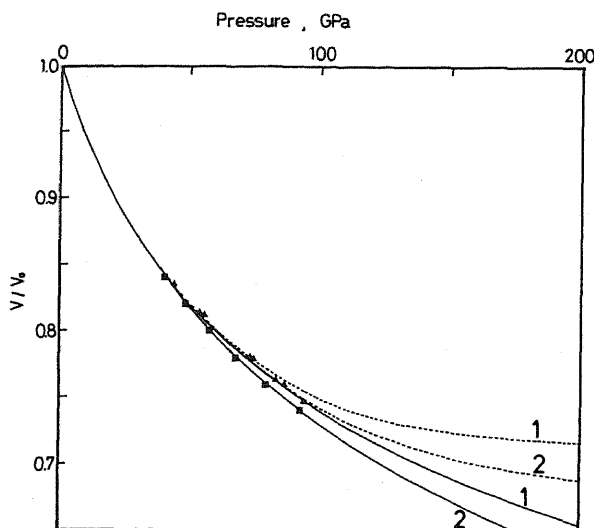


Fig. 8. Isothermal compression curves of MgO at 300 K. Both the Birch-Murnaghan EOS (solid line) and the shock-EOS (dotted line) are shown for #1 and #2 in Table 4. The data of static compression in DAC (Mao and Bell, 1979; solid triangle) and the shock-EOS calculated by Jamieson *et al.* (1982; solid square) are also shown for comparison.

wurtzite-structured AgI; negative thermal expansion is observed from 120 to 300 K under atmospheric pressure (Bienstock and Barley, 1963). Fjeldly and Hanson (1974) observed negative signs of C'_{11} , C'_{44} and C'_{66} , and suggested that these softenings of elastic constants are associated with negative thermal expansion.

4.3 Elastic properties of MgAl_2O_4

Table 9 compiles the τ'_{obs} and the τ'_{cal} s for MgAl_2O_4 , where the pressure values by MgO sensor are based on #1 results of MgO; even if we use the #2 results for MgO sensor, the difference of pressure values is small enough (<0.03 GPa) to neglect. The analysis was made in a model involving both ξ and η ; the resultant pressure derivatives are shown in Table 10, together with those by Chang and Barsch (1973). Considering the errors of η , a model involving $\eta \equiv 0$ for all elastic constants appears to be inappropriate for MgAl_2O_4 in contrast with MgO. Significant discrepancies are found between the present results and those by Chang and Barsch; all the first derivatives, K'_S , C'_S , and C'_{44} , are inconsistent with those by Chang and Barsch beyond the uncertainty. In particular, the present C'_S is negative against their positive C'_S . The slight increase of C_S in their study does not seem to be an intrinsic property of MgAl_2O_4 , because the overall magnitude of increase of C_S with pressure in their measurement is almost the same as that of the experimental uncertainty. The decrease of the travel time in their study may be caused by the deficit of the bond correction.

The pressure dependence of elastic constants is shown in Fig. 9. Not only C_S but also K_S and C_{44} are expected to show softening owing to the definitely negative second

Table 8. Pressure-volume-temperature relation of #1 shock-EOS of MgO. Pressure is in GPa. Normalized volume of 1.0 corresponds to density of 3,584 kg/m³.

Volume	Temperature (K)							
	100	300	500	700	900	1,100	1,300	1,500
1.00	−0.64(GPa)	0.01	1.07	2.26	3.49	4.75	6.02	7.30
0.99	1.04	1.66	2.69	3.85	5.06	6.28	7.53	8.78
0.98	2.81	3.41	4.41	5.53	6.71	7.91	9.12	10.34
0.97	4.68	5.25	6.22	7.31	8.46	9.62	10.80	11.99
0.96	6.65	7.20	8.13	9.19	10.31	11.44	12.59	13.75
0.95	8.74	9.26	10.16	11.19	12.26	13.36	14.48	15.60
0.94	10.94	11.43	12.30	13.29	14.33	15.40	16.47	17.56
0.93	13.27	13.73	14.56	15.52	16.52	17.55	18.59	19.64
0.92	15.74	16.17	16.96	17.87	18.83	19.82	20.82	21.83
0.91	18.34	18.75	19.49	20.36	21.28	22.23	23.18	24.15
0.90	21.11	21.48	22.18	23.00	23.88	24.77	25.68	26.60
0.89	24.03	24.37	25.03	25.80	26.62	27.47	28.32	29.19
0.88	27.14	27.44	28.05	28.76	29.53	30.32	31.12	31.93
0.87	30.45	30.71	31.25	31.91	32.61	33.34	34.08	34.83
0.86	33.96	34.18	34.67	35.25	35.89	36.54	37.21	37.89
0.85	37.72	37.89	38.30	38.81	39.36	39.94	40.53	41.13
0.84	41.73	41.85	42.18	42.60	43.07	43.55	44.05	44.56
0.83	46.04	46.10	46.33	46.66	47.02	47.40	47.79	48.20
0.82	50.69	50.67	50.79	51.00	51.24	51.50	51.77	52.05
0.81	55.72	55.61	55.61	55.67	55.77	55.89	56.02	56.15
0.80	61.20	60.99	60.83	60.72	60.65	60.60	60.55	60.51
0.79	67.24	66.89	66.53	66.22	65.94	65.67	65.40	65.15
0.78	73.97	73.44	72.83	72.26	71.71	71.17	70.63	70.10
0.77	81.62	80.83	79.90	78.99	78.08	77.18	76.29	75.39
0.76	90.54	89.40	88.01	86.62	85.23	83.84	82.46	81.07
0.75	101.43	99.73	97.67	95.58	93.48	91.37	89.27	87.17
0.74	115.76	113.10	109.93	106.70	103.45	100.19	96.93	93.66
0.73	137.64	133.00	127.67	122.23	116.76	111.28	105.79	100.29

pressure derivatives. The large negative η_{ks} and η_{cs} for MgAl₂O₄ may be related to the decomposition to oxides at 14 GPa and further recombination to a new phase at 25 GPa (Liu, 1975, 1978, 1980; Ohtani *et al.*, 1975).

4.4 Pressure derivative of rigidity of γ -Mg₂SiO₄

The γ -Mg₂SiO₄ is regarded as an important candidate mineral in the layer of 500–670 km depth in the mantle (e.g., Akimoto *et al.*, 1976). The elastic properties of γ -Mg₂SiO₄ may be estimated from the data of MgAl₂O₄, because MgAl₂O₄ has the same structure Fd3m with γ -Mg₂SiO₄. Tables 11 and 12 compare physical properties of MgAl₂O₄ and γ -Mg₂SiO₄.

The pressure derivative of rigidity, μ' , of γ -Mg₂SiO₄ has not been measured yet, although it is important to interpret the seismic wave profiles of the upper mantle. The μ' of MgAl₂O₄ aggregate (−0.13–0.84, Table 12) may be assumed for that of γ -Mg₂SiO₄. However, such simple analogy may be misleading, because there are several differences

Table 9. Comparison between the observed travel times, τ_{obs} , and the calculated travel times, τ_{cal} , for MgAl_2O_4 , combined with pressure values. τ_{obs} is the measured travel time after the corrections.

[100]P (3.606 mm)			[110]P (3.910 mm)		
P (GPa)	τ_{obs} (ns)	τ_{cal} (ns)	P (GPa)	τ_{obs} (ns)	τ_{cal} (ns)
3.44	792.0	791.8	2.58*	751.4	751.0
4.55	787.4	787.8	2.74*	750.7	750.4
5.36	785.5	785.5	0.99	757.8	758.9
3.74	791.7	790.6	2.77	749.4	750.3
			4.23	744.6	745.1

[100]S (3.604 mm)			[110]S (3.969 mm)		
P (GPa)	τ_{obs} (ns)	τ_{cal} (ns)	P (GPa)	τ_{obs} (ns)	τ_{cal} (ns)
2.54	1,087.3	1,087.4	2.58*	1,885.9	1,886.3
4.54	1,084.9	1,083.5	2.74*	1,886.4	1,886.7
6.16	1,081.5	1,082.0	7.78*	1,900.0	1,900.0

* Pressures are determined by means of bismuth calibrant. The others are by means of MgO pressure sensor based on #1 results in Table 4.

Table 10. Results for MgAl_2O_4 in comparison with those by Chang and Barsch (1973).

Present work	Mode					
	K_S	C_S	C_{44}	C_{11}	C_{12}	K_T
C_0 (GPa)	197.9 ± 0.2	63.8 ± 0.1	154.8 ± 0.2	282.9	155.4	196.4
C'	5.66 ± 0.21	-0.049 ± 0.03	1.44 ± 0.10	5.59	5.69	5.68
C'' (GPa^{-1})	-0.65 ± 0.10	-0.0071 ± 0.01	-0.19 ± 0.04	-0.65	-0.64	-0.64
ξ_C	1.62 ± 0.21	-4.15 ± 0.10	-2.17 ± 0.13	-0.12	3.19	1.68
η_C	-117 ± 19	11.8 ± 6.1	-33.7 ± 10.5	-78.6	-152.8	-117
Chang and Barsch						
C_0	197.4	63.8	154.7	282.5	154.9	195.7
C'	5.15	0.1952	0.89	5.15	4.76	
C''	-0.48	-0.029	-0.029	-0.52	-0.46	

in structure and elastic property between them, such as u -parameter and anisotropy factor (C_S/C_{44}).

I propose an alternative method to estimate μ' of $\gamma\text{-Mg}_2\text{SiO}_4$. As shown in Eq. (C.12), the pressure derivative of elastic constant is connected with the corresponding mode Grüneisen parameter. In the following, a method is developed to estimate the mode Grüneisen parameters of $\gamma\text{-Mg}_2\text{SiO}_4$ from those of MgAl_2O_4 by means of lattice dynamic analogy.

Averaged Grüneisen parameter of acoustic branch, γ_{ac} of $\gamma\text{-Mg}_2\text{SiO}_4$ is calculated by use of the relation:

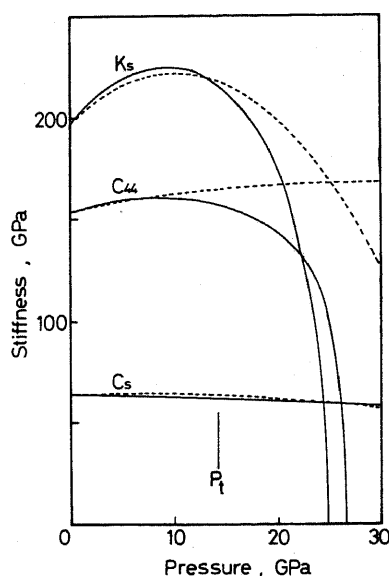


Fig. 9. Extrapolation of stiffness of MgAl_2O_4 based on the present results (solid line). Dashed lines correspond to those by Chang and Barsch (1973). All the stiffnesses show softening over 10 GPa. P_t : transition pressure.

Table 11. Physical parameters of MgAl_2O_4 and $\gamma\text{-Mg}_2\text{SiO}_4$. Some data for MgAl_2O_4 are already shown in Table 2. The u -parameter for regularly octahedral packing is 0.375.

	MgAl_2O_4	$\gamma\text{-Mg}_2\text{SiO}_4$
Lattice constant (pm)	808.36 ^a	806.5 ^c
u -Parameter	0.3883 ^b	0.3685 ^c
Density (kg/m ³)	3,578.0	3,562
Specific heat C_p (J/kg K)	815	765.9 ^d
Thermal expansion β (K ⁻¹)	1.9×10^{-5}	1.87×10^{-6} ^e
Grüneisen constant γ	1.29	1.26

^a Present data. ^b Finger *et al.*, 1986. ^c Sasaki *et al.*, 1982. ^d Watanabe, 1982. ^e Suzuki *et al.*, 1979.

$$\gamma_{ac} = (C_l \gamma_l + C_s \gamma_s) / C_{ac}, \quad (16)$$

where sub-suffix “ac” refers to total acoustic branches, and those of “l” and “s” refer to longitudinal acoustic branch and transverse one, respectively; γ is mode Grüneisen parameter, and C is mode heat capacity. Using Eq. (C.12), the γ_l and γ_s are given by pressure derivatives of elastic constant;

$$\gamma_l = K_T(\lambda' + 2\mu') / 2(\lambda + 2\mu) - 1/6 \quad (17)$$

$$\gamma_s = K_T\mu' / 2\mu - 1/6, \quad (18)$$

where λ and μ are Lamé's elastic constants in isotropic elasticity. Substituting Eqs. (17)

Table 12. Elastic constants and their pressure derivatives of MgAl_2O_4 and $\gamma\text{-Mg}_2\text{SiO}_4$. $\lambda+2\mu$ and μ are elastic constants for longitudinal and shear waves of isotropic aggregate, respectively; the lower bound and the upper one correspond to the Reuss and the Voigt averages.

	MgAl_2O_4	$\gamma\text{-Mg}_2\text{SiO}_4$
K_S (GPa)	197.9	183.7*
K'_S	5.66	—
K_T (GPa)	196.4	182.4
K'_T	5.68	4.1–4.9**
C_S (GPa)	63.8	107.5 ^a
C'_S	−0.049	—
C_{44} (GPa)	154.8	126.0 ^a
C'_{44}	1.44	—
$\lambda+2\mu$ (GPa)	329.3–355.7	340.9–341.8
$(\lambda+2\mu)'$	5.49–6.78	—
μ (GPa)	98.5–118.4	117.9–118.6
μ'	−0.13–0.84	—

* Weidner *et al.*, 1984. ** Sawamoto *et al.*, personal communication.

and (18) into Eq. (16), we have γ_{ac} as a function of λ' and μ' . On the other hand, Sawamoto *et al.* (personal communication) obtained 4.1 ± 0.8 for K'_T ($=\lambda'+2\mu'/3$) of $\gamma\text{-Mg}_2\text{SiO}_4$ by means of X-ray diffraction measurement under hydrostatic compression. Note that γ_{ac} and K'_T are independent functions of λ' and μ' . Therefore we can obtain λ' and μ' if we have a value of γ_{ac} of $\gamma\text{-Mg}_2\text{SiO}_4$.

The γ_{ac} of $\gamma\text{-Mg}_2\text{SiO}_4$ is estimated from the elasticity data of MgAl_2O_4 . The γ_{ac} of MgAl_2O_4 is calculated as 0.83 by use of the equation

$$\gamma_{ac} = \sum \gamma_i C_i / \sum C_i, \quad (19)$$

where i ($i=1-4$) corresponds to the four modes in Eq. (C.1). Averaged Grüneisen parameter of optic modes, γ_{op} , of MgAl_2O_4 is calculated to be 1.34 by use of the equation

$$\gamma = (C_{ac}\gamma_{ac} + C_{op}\gamma_{op}) / C_v, \quad (20)$$

where γ (Grüneisen constant) is given by $V\beta K_T/C_v$, and C_{op} (mode heat capacity for optic branches) by $C_v - C_{ac}$.

The γ_{ac} of $\gamma\text{-Mg}_2\text{SiO}_4$ may be estimated by means of the following two typical assumptions: (1) The acoustic properties of $\gamma\text{-Mg}_2\text{SiO}_4$ are analogous to that of MgAl_2O_4 . Thus 0.83 is assumed for the γ_{ac} of $\gamma\text{-Mg}_2\text{SiO}_4$. (2) The averaged properties of optic branches of $\gamma\text{-Mg}_2\text{SiO}_4$ are analogous to that of MgAl_2O_4 , as a result of average of 36 optic modes (O'horo *et al.*, 1973). Thus 1.34 is assumed for the γ_{op} of $\gamma\text{-Mg}_2\text{SiO}_4$, and 0.61 is calculated for the γ_{ac} of $\gamma\text{-Mg}_2\text{SiO}_4$ by using Eq. (20).

Thus we can reasonably take the range of 0.61–0.83 for the γ_{ac} of $\gamma\text{-Mg}_2\text{SiO}_4$. The K'_S of $\gamma\text{-Mg}_2\text{SiO}_4$ is taken to be 4.1–4.9, because $K'_S < 4.1$ is quite unlikely for minerals. Using these estimated ranges we obtain 0.56–1.0 as a possible range of μ' for $\gamma\text{-Mg}_2\text{SiO}_4$, which corresponds to -3.1 – -2.4 of $\xi_\mu (=K_T\mu'/\mu-4)$ of $\gamma\text{-Mg}_2\text{SiO}_4$. Figure 10 shows the extrapolation of K and μ of $\gamma\text{-Mg}_2\text{SiO}_4$. It is remarkable that the lower marginal

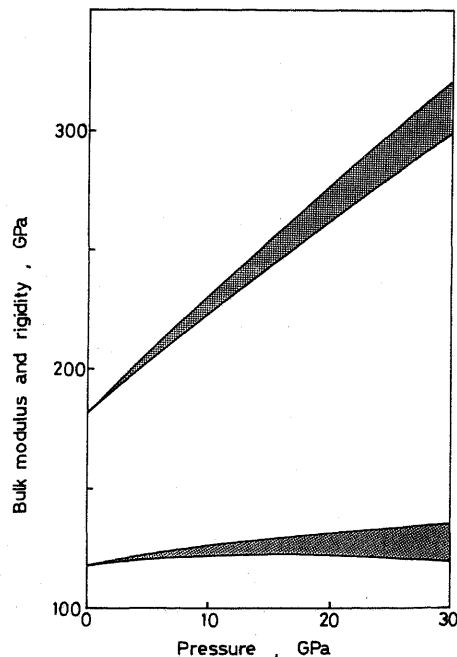


Fig. 10. Extrapolation of bulk modulus K and rigidity μ of γ - Mg_2SiO_4 by use of the Birch-Murnaghan equations.

value of ξ_μ provides shear softening of γ - Mg_2SiO_4 at pressure around 20 GPa.

5. Some Geophysical Implications

5.1 Softening of shear constant in minerals

The present study implicates that softening of shear constant in minerals is expected more frequently than has been expected. The shear softening is a possible explanation for shear wave anomaly (negative or positively low gradient of shear wave velocity) in the mantle, instead of current explanations such as large temperature gradient, compositional variation, and mineralogical variation in the corresponding layer.

In some seismological velocity models, the shear wave anomalies have been suggested for the layer of 500–670 km depth (e.g., Ibrahim and Nuttli, 1967; Anderson and Julian, 1969; Jordan and Anderson, 1974) and for the D'' layer just above the mantle core boundary (e.g., Doornbos and Mondt, 1979; Lay and Helmberger, 1983). As for the anomaly in the D'' layer, Garnero *et al.* (1988) reported the lateral heterogeneity involving even both positive and negative slopes of shear velocity.

γ - Mg_2SiO_4 is expected as a major constituent of the layer between 500 and 670 km depth in the mantle. Therefore we can imagine the shear velocity anomaly in this layer caused by softening of rigidity of γ - Mg_2SiO_4 (Fig. 10).

MgO is a candidate constituent of the lower mantle (e.g., Liu, 1977; Ito and Matsui, 1978). Note that MgO has a softening shear constant (C_{44}) and a hardening shear

constant (C_s). The anisotropy of shear constant in MgO can yield the lateral heterogeneity in the layer through preferred orientation of MgO crystals.

5.2 Negative thermal expansion

Positive thermal expansion has been presupposed in the study of mantle dynamics. However, this presupposition must be discarded because of a possibility of negative thermal expansion of MgO suggested in the present study. We must take account of the possibility of negative thermal expansion under high pressure for other oxide and silicate minerals.

The negative thermal expansion of MgO is associated with softening of C_{44} with pressure. In general, negative thermal expansion can occur in association with decrease of some frequencies of lattice vibration with pressure. In addition to data of pressure dependence of elastic constants corresponding to acoustic branches, information of pressure effect on optical branches is needed to predict the effect of pressure on thermal expansivity of minerals.

A direct effect of negative thermal expansion is to suppress thermal convection. The layer of negative thermal expansion may behave as a barrier of thermal convection. Therefore it may be a thermal boundary layer, where only the thermal conduction is responsible for heat transport. The layer can behave also as a chemical insulator between the upper and the lower layers, because the material transport is only by diffusion process which is usually too slow to mix the materials in the two layers. In other words, the layer may be a strong barrier against convectational mixing of the mantle.

6. Conclusions

(1) Technical development was performed for pulse-echo-overlap method at high hydrostatic pressure. In addition to the homogeneous metal gaskets and the liquid-solid hybrid system (Kozuki *et al.*, 1986; Yoneda, 1987), various new devices such as the indium bond and buffer-rod of hardened steel facilitated accurate measurement of the acoustic velocity for large single crystal sample up to ~ 8 GPa.

(2) Analytical procedure of acoustic data processing was improved in the two items. A self-consistent correction method was developed in order to obtain precise travel time. A new formulation was developed to describe pressure dependence of elastic constants.

(3) Elastic constants and their first and second pressure derivatives of MgO and MgAl_2O_4 were determined. Extrapolations of the elastic constants to higher pressure show that the pressure-induced shear softening is quite probable in both minerals. The relation of phase transition and shear softening in MgO and MgAl_2O_4 is discussed by means of Born's criterion.

(4) Isothermal EOS of MgO was examined in various methods. Physical properties (heat capacity, Grüneisen constant, thermal expansion, etc.) of MgO were calculated by modifying the Kieffer model. Negative thermal expansion is expected for MgO above ~ 50 GPa.

(5) The possible range of pressure derivatives of rigidity in $\gamma\text{-Mg}_2\text{SiO}_4$ was estimated from the present data of MgAl_2O_4 ; the lower value indicates the shear softening

of γ - Mg_2SiO_4 at the pressures around 20 GPa.

(6) Geophysical implications of shear softening of MgO and γ - Mg_2SiO_4 , and negative thermal expansion of MgO are discussed.

I am indebted to professors Yasuo Shimazu, Yoshio Fukao, and Hiroshi Sawamoto, who supervised my research activities. I would like to thank Mineo Kumazawa of the University of Tokyo, and Hitoshi Mizutani of the Institute of Space and Astronautical Science for their discussion on the present work. I am grateful to Dr. Hideyuki Fujisawa of the University of Tokyo for discussion on ultrasonic technique, to Messrs. Gunzo Takamatsu, Tadashi Masuda, Syuzo Isikawa, Kazuji Koizumi, Tatsuharu Torii, and Miss Chiyomi Miwa of the Instrument Development Center of Faculty of Science, Nagoya University, for their help and advice during technical development, and to Messrs. Isuzu Hiraiwa and Setuo Yogo for their help and advice in sample and transducer preparation. I thank Drs. Manabu Kato, Akio Fujimura, Yoshio Sumino, Isao Yamada, and other colleagues for their advice and discussions. The early stage of the present work was done in cooperation with Dr. Yasushi Kozuki. The manuscript was substantially improved through critical readings by M. Kumazawa and by the anonymous reviewers.

Computation was carried out by FACOM M780/20 at the Computation Center of Nagoya University (No. A40739A).

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APPENDIX

A. Corrections to Measured Travel Time

1. Correction of diffraction effect

Diffraction effect in the ultrasonic field gives rise to a phase shift of observed echo pulse (e.g., Seki *et al.*, 1956). The nature of diffraction is explained in Fig. A1 in terms of point-source spherical wave. The amount of phase difference, D , between the observed signal and the imaginary plane wave is calculated as follows.

The amplitude of signal detected by a receiver $a(t, z)$ (t , time; z , positional coordinate

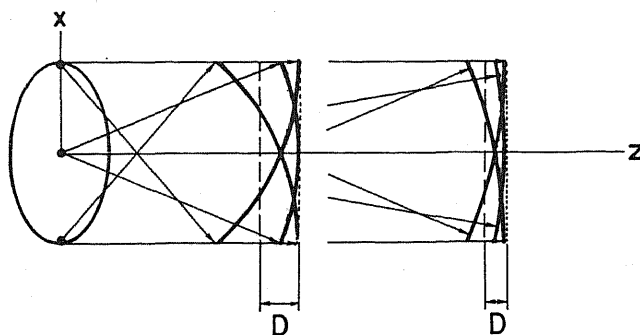


Fig. A1. Point-source spherical wave fronts emitted from the central point and two opposite points on edge of a radiating transducer are illustrated by bold curves as representatives. The envelope for the spherical waves shown by dotted line corresponds to the imaginary plane wave constituted only by wave elements radiated perpendicularly. The real wave is a synthesis of all possible spherical waves. Its phase, given as an average of those waves, inevitably delays from that of the imaginary plane wave. Suppose the dashed line represents the averaged phase, the diffraction phase shift, D , is the difference between the dotted line and the dashed line. Note that the scale of x coordinate is much magnified in comparison with that of z coordinate in order to emphasize the effect. It is evident that the diffraction phase shift decreases with increasing z .

shown in Fig. A1) is represented by

$$a(t, z) = -a_0 \int \exp\{i(\omega t - kr)\} / r \, ds_1 ds_2, \quad (\text{A.1})$$

where a_0 is proportional coefficient depending on experimental condition; ω , angular frequency; k , ($=\omega/\sqrt{(C/\rho)}$) wave number; r , distance between a point on radiator and that on receiver; ds_1 , infinitesimal surface element of radiator and ds_2 that of receiver. Equation (A.1) is modified by subtracting plane wave component $\exp\{i(\omega t - kz)\}$;

$$a(t, z) = -a_0 \exp\{i(\omega t - kz)\} \int \exp(-ikd) / r \, ds_1 ds_2, \quad (\text{A.2})$$

where $d (=r-z)$ denotes excess path length.

Equation (A.2) is modified to

$$a(t, z) = -a \exp\{i(\omega t - kz - D)\}, \quad (\text{A.3})$$

where the phase shift D and the amplitude a for the synthesized plane wave are calculated by the numerical integration.

2. Correction of bond effect

The echo pulses I, L, M (Fig. 4) are composite of multiple wave elements ($I_0, I_1, I_2, \dots; L_1, L_2, L_3, \dots; M_1, M_2, M_3, \dots$), arising from multiple reflections at both

ends of bond, where subscript numerals refer to the times of round-trip of acoustic wave through bond. The "bond phase shift" is defined as difference of phases between the first wave element (I_0 , L_1 , M_1) and the observed wave, which is composite of all wave elements. The synthetic waves for signals I, L, M are

$$I_0 + I_1 + I_2 + \cdots = A_I \exp\{i(\omega t - kz - D_I - B_I)\} \quad (\text{A.4})$$

$$L_1 + L_2 + L_3 + \cdots = A_L \exp\{i[\omega(t - \tau) - kz - \Delta - D_L - B_L]\} \quad (\text{A.5})$$

$$M_1 + M_2 + M_3 + \cdots = A_M \exp\{i[\omega(t - 2\tau) - kz - \Delta - D_M - B_M]\} \quad (\text{A.6})$$

where A_I , A_L , A_M are amplitudes, B_I , B_L , B_M phase shifts due to bond effect, and D_I , D_L , D_M phase shifts due to diffraction effect; τ is the intrinsic travel time in sample; and $\Delta (= \omega\tau_b$; τ_b : travel time of wave through a round-trip in bond) is the phase delay of wave due to bond. We have to consider the effect of the dissipation factor G in bond material on amplitude (A) and bond phase shift (B), because the energy dissipation in indium is found to be quite large. The expression of each wave element is shown in Table A1.

Table A1. Formulation of each wave element involving diffraction and bond effects. Successive wave elements beyond the third one are not shown because of their complicated expressions. T_1 and R_1 are transmission and reflection coefficients, respectively, of wave energy at the interface between buffer-rod and bond; T_2 and R_2 , those at the interface between bond and sample; and R_3 , reflection coefficient at the other end of sample.

$$I_0 = a_I \sqrt{(R_1)} \exp\{i(\omega t - kz - D_I)\}$$

$$I_1 = -a_I \sqrt{(T_1 R_2 T_1)} G \exp\{i(\omega t - kz - \Delta - D_I)\}$$

$$I_2 = -a_I \sqrt{(T_1 R_2 R_1 R_2 T_1)} G^2 \exp\{i(\omega t - kz - 2\Delta - D_I)\}$$

...

$$L_1 = a_L \sqrt{(T_1 T_2 R_3 T_2 T_1)} G \exp\{i[\omega(t - \tau) - kz - \Delta - D_L]\}$$

$$L_2 = 2a_L \sqrt{(T_1 R_2 R_1 T_2 R_3 T_2 T_1)} G^2 \exp\{i[\omega(t - \tau) - kz - 2\Delta - D_L]\}$$

$$L_3 = 3a_L \sqrt{(T_1 R_2 R_1 R_2 R_1 T_2 R_3 T_2 T_1)} G^3 \exp\{i[\omega(t - \tau) - kz - 3\Delta - D_L]\}$$

...

$$M_1 = a_M \sqrt{(T_1 T_2 R_3 R_3 T_2 T_1)} G \exp\{i[\omega(t - 2\tau) - kz - \Delta - D_M]\}$$

$$M_2 = a_M \{2\sqrt{(T_1 R_2 R_1 T_2 R_3 R_3 T_2 T_1)} - \sqrt{(T_1 T_2 R_3 T_2 R_1 T_2 R_3 T_2 T_1)}\} G^2 \exp\{i[\omega(t - 2\tau) - kz - 2\Delta - D_M]\}$$

$$M_3 = a_M \{3\sqrt{(T_1 R_2 R_1 R_2 R_1 T_2 R_3 R_3 T_2 T_1)} - 3\sqrt{(T_1 R_2 R_1 T_2 R_3 T_2 R_1 T_2 R_3 T_2 T_1)}\} G^3 \\ \times \exp\{i[\omega(t - 2\tau) - kz - 3\Delta - D_M]\}$$

...

Reflection coefficient R and transmitting coefficient T of wave energy at interface between different substance are given by $R = (Z_1 - Z_2)^2 / (Z_1 + Z_2)^2$; $T = 4Z_1 Z_2 / (Z_1 + Z_2)^2$, where Z_1 and Z_2 are acoustic impedance of the two substances, respectively. Note that phase of reflecting wave changes with 0 or π radian depending on relative magnitude between Z_1 and Z_2 . In the present work, Z of indium bond is smaller than those of the buffer-rod and the present samples. The effect of indium film is considered by a factor G , which denotes an attenuation coefficient to one round-trip of the acoustic wave.

The problem is reduced to find a set of three independent parameters τ , A , and G under the two types of constraints:

(1) the relative magnitude among signals:

$$(A_L/A_I)_o = (A_L/A_I)_s \quad (\text{A.7})$$

$$(A_M/A_L)_o = (A_M/A_L)_s \quad (\text{A.8})$$

where subscripts "o" and "s" refer to observed and synthetic quantities, respectively;

(2) the observed travel time:

$$(T_{IL})_o = \tau + w^{-1}(\Delta + B_L + D_L - B_I - D_I) \quad (\text{A.9})$$

$$(T_{LM})_o = \tau + w^{-1}(B_M + D_M - B_L - D_L). \quad (\text{A.10})$$

The numerical computation is made by iterative method with several times of cycle. The wave elements with small amplitude (less than 10% of the first wave) is omitted in the iteration, because they hardly affect the amount of correction.

3. On elastic data used in the correction

The elastic constants and their pressure dependence of sample, buffer-rod, liquid medium, and bond are necessary to make the present correction for high pressure data.

For sample, the uncorrected elastic data is sufficient for the present purpose. For buffer-rod, the elastic constants of armco iron by Hughes and Maurette (1956) was used by means of linear extrapolation. For liquid medium, the velocity of longitudinal wave was estimated from the pressure-volume relation of methyl alcohol reported by Bridgman (1942).

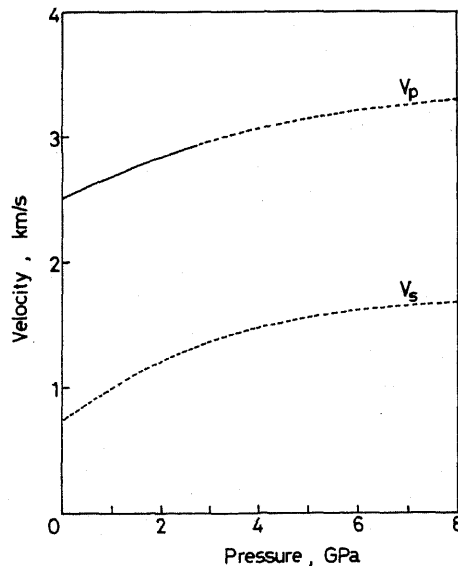


Fig. A2. Pressure dependence of acoustic wave velocity in indium. Solid line is the result of the experiment; dotted line, an estimated result described in the text.

For indium, we have not sufficient data for the present purpose. The data available for indium were elastic constant of single crystal at atmospheric pressure (Winder and Smith, 1958) and compression curve to ~ 10 GPa (Bridgman, 1941). The former data were converted to those of isotropic aggregate by means of Voigt-Reuss-Hill averaging method (Hill, 1952), and the latter data were revised by converting the old pressure scale to the current one. Further, the longitudinal wave velocity of polycrystalline indium was measured to 2.74 GPa (low Bi point), while the measurement of transverse velocity failed because of extremely high dissipation in indium (~ 1 mm thickness). Combining all the data available, the acoustic velocity in indium was evaluated as shown in Fig. A2.

B. Derivation of Formulas for EOS

Equations (5)–(10) are derived as follows. In the Birch-Murnaghan EOS, the elastic energy Φ is described by the finite strain ε ;

$$\Phi = K_{T_0} \{ \varepsilon^2/2 + (\xi_{k_T}/6)\varepsilon^3 + (\eta_{k_T}/24)\varepsilon^4 + \dots \} \quad (\text{B.1})$$

$$\varepsilon = (3/2)(V^{-2/3} - 1), \quad (\text{B.2})$$

where V is normalized volume, ξ_{k_T} and η_{k_T} non-dimensional parameters. Equation (B.2) is the definition of ε . Pressure and isothermal bulk modulus are given as

$$\begin{aligned} P &= -\partial\Phi/\partial V = -(\partial\Phi/\partial\varepsilon)(\partial\varepsilon/\partial V) \\ &= K_{T_0}(2\varepsilon/3 + 1)^{5/2} \{ \varepsilon + (\xi_{k_T}/2)\varepsilon^2 + (\eta_{k_T}/6)\varepsilon^3 \} \end{aligned} \quad (\text{B.3})$$

$$\begin{aligned} K_T &= -V(\partial P/\partial V) = K_{T_0}(2\varepsilon/3 + 1)^{5/2} \{ 7\varepsilon/3 + 1 + (\xi_{k_T}/2)\varepsilon(9\varepsilon/3 + 2) \\ &\quad + (\eta_{k_T}/6)\varepsilon^2(11\varepsilon/3 + 3) \}. \end{aligned} \quad (\text{B.4})$$

K'_T and K''_T are described as

$$K'_T = \partial K_T / \partial P \big|_{P=0} = \xi_{k_T} + 4 \quad (\text{B.5})$$

$$K''_T = \partial K'_T / \partial P \big|_{P=0} = K_{T_0}^{-1} (\eta_{k_T} - \xi_{k_T}^2 - \xi_{k_T} - 35/9). \quad (\text{B.6})$$

In analogy with Eq. (B.4), the pressure dependence of an elastic stiffness C should be described as

$$C = C_0(2\varepsilon/3 + 1)^{5/2} \{ 7\varepsilon/3 + 1 + (\xi_C/2)\varepsilon(9\varepsilon/3 + 2) + (\eta_C/6)\varepsilon^2(11\varepsilon/3 + 3) \} \quad (\text{B.7})$$

C' and C'' are expressed as follows from Eqs. (B.3) and (B.7);

$$C' = \partial C / \partial P \big|_{P=0} = (C_0/K_{T_0})(\xi_C + 4) \quad (\text{B.8})$$

$$\begin{aligned} C'' &= \partial C' / \partial P \big|_{P=0} \\ &= (C_0/K_{T_0}^2)(\eta_C - \xi_{k_T}\xi_C - 4\xi_{k_T} + 3\xi_C - 35/9). \end{aligned} \quad (\text{B.9})$$

From Eqs. (B.8) and (B.9), we have

$$\xi_C = (K_{T_0}/C_0)C' - 4 \quad (\text{B.10})$$

$$\eta_C = (K_{T_0}^2/C_0)C'' + (K_{T_0}/C_0)K_T' C' - 7(K_{T_0}/C_0)C' + 143/9, \quad (\text{B.11})$$

where Eqs. (B.10) and (B.11) are identical with Eqs. (9) and (10), respectively. We have Eqs. (7) and (8) by substituting K_T into C in Eqs. (B.10) and (B.11), or rewriting Eqs. (B.5) and (B.6) with respect to ξ_{k_T} and η_{k_T} .

Suppose that a stiffness C is described as

$$C = \sum_i a_i C_i, \quad (\text{B.12})$$

where a_i is conversion coefficient between C and C_i . Then, ξ_C and η_C are described as

$$\xi_C = (\sum_i a_i C_{i0} \xi_{C_i}) / C_0 \quad (\text{B.13})$$

$$\eta_C = (\sum_i a_i C_{i0} \eta_{C_i}) / C_0. \quad (\text{B.14})$$

The other topic in derivation of formula is the pressure dependence of κ in Eq.(12). Its pressure derivative is

$$\begin{aligned} (\partial\kappa/\partial P)_T &= -\kappa^2(V\beta^2 K_S T / C_p) \\ &\times \{ -\kappa^{-1} K_S^{-1} + 2\beta^{-1}(\partial\beta/\partial P)_T + K_S^{-1}(\partial K_S/\partial P)_T - C_p^{-1}(\partial C_p/\partial P)_T \}. \end{aligned} \quad (\text{B.15})$$

The first term and the third one in the brace can be estimated from the present experimental results. $(\partial\beta/\partial P)_T$ in the second term and $(\partial C_p/\partial P)_T$ in the fourth one have not been measured yet experimentally. However, Overton (1962) proposed to estimate them from the experimental results of $(\partial K_S/\partial T)_P$ and $(\partial\beta/\partial T)_P$ by means of the following thermodynamic relations, Eqs. (B.16)–(B.18). The $(\partial\beta/\partial P)_T$ in the second term is rewritten as

$$\begin{aligned} (\partial\beta/\partial P)_T &= K_T^{-2}(\partial K_T/\partial T)_P \\ &= (\kappa K_S)^{-2} \{ \kappa(\partial K_S/\partial T)_P + K_S(\partial\kappa/\partial T)_P \} \\ &= (\kappa K_S)^{-2} [\kappa(\partial K_S/\partial T)_P - K_S \kappa^2 \gamma \{ \beta + T(\partial\beta/\partial T)_P \}], \end{aligned} \quad (\text{B.16})$$

where

$$(\partial\kappa/\partial T)_P = -\kappa^2 \gamma \{ \beta + T(\partial\beta/\partial T)_P \} \quad (\text{B.17})$$

is used to obtain the final expression of Eq. (B.16); Eq. (B.17) is derived under the assumption that the Grüneisen constant is independent of temperature, i.e., $(\partial\gamma/\partial T)_P \equiv 0$. The $(\partial C_p/\partial P)_T$ in the fourth term is expressed exactly as

$$\begin{aligned} (\partial C_p/\partial P)_T &= T(\partial^2 S/\partial P \partial T) = -T(\partial^2 V/\partial T^2)_P \\ &= -TV \{ \beta^2 + (\partial\beta/\partial T)_P \}. \end{aligned} \quad (\text{B.18})$$

C. Modified Kieffer Model

Kieffer (1979a, b, c, 1980, 1982) proposed a new model for estimating heat capacity of mineral, and demonstrated its superiority over the Debye model for a number of minerals. Kieffer assumed sine function for dispersion curve of acoustic branch, and homogeneous distribution of mode density for optic branch. The Kieffer model requires

Table C1. Heat capacity C_v at 1 bar, 300 K for MgO and NaCl.

	Experiment	Debye model			Kieffer model	Modified Kieffer model	
		Voigt	Hill	Reuss		Model A	Model B
MgO	36.7 (J/mol·K)	31.62	31.90	32.17	35.5	35.99	36.07
NaCl	48.5	47.34	47.38	47.41	47.3	47.84	47.83

the optical data (Raman scattering, infrared absorption, and/or inelastic neutron diffraction) in order to determine the frequency bound of optic branch. It is usually difficult to estimate heat capacity under high pressure by means of the Kieffer model because of insufficient experimental data. On the other hand, the pressure dependence of heat capacity is easily estimated by applying the Debye model to experimental results of pressure dependence of acoustic velocity. In other words, the Kieffer model is more accurate but it is not usually applicable under high pressure in contrast with the Debye model.

For minerals having two different atoms in a primitive unit cell such as alkali halides, a good compromise appears to be to modify the Kieffer model by using a theoretical relation between acoustic branch and optic branch (e.g., Kittel, 1974). In modifying the Kieffer model, heat capacity is calculated separately for longitudinal and transverse modes by use of the theoretical bounds instead of the experimental bounds. For this reason, pressure effect on both acoustic and optic branches can be estimated by use of the data of acoustic velocity.

In an anisotropic crystal, it is difficult to determine average velocity; Kieffer uses the Voigt-Reuss-Hill average for longitudinal velocity and an ad hoc method to determine two transverse velocities. In the modified Kieffer model, I use the Reuss average for longitudinal velocity, and examine the following two methods for transverse velocities: (A) two transverse velocities corresponding to C_{44} and C_S with weight ratio of 3:2 as analogy to the Voigt and the Reuss average methods; and (B) three transverse velocities corresponding to C_{44} , C_S , and C_{111} [$= (2C_S + C_{44})/3$: Shear stiffness along (111) direction]. For cubic crystal, the weight ratio of 6:3:4 among C_{44} , C_S , and C_{111} are determined by taking account of the number of the crystal direction (6 of [100]; 12 of [110]; 8 of [111]), and mode degeneracy of shear mode in [100] and [111] directions.

Table C1 compiles the observed and calculated values of heat capacity, C_v , of MgO and NaCl under the standard state. It is evident that the modified Kieffer models are superior to both the Debye model and the original Kieffer model, and the discrepancies between the model A and the model B is very small. In the present work, I use the model B for analysis of physical properties of MgO. Note that the modified Kieffer model cannot be used for MgAl_2O_4 because it has as many as 14 atoms ($2\text{MgAl}_2\text{O}_4$) in a primitive unit cell.

The molar heat capacity at constant volume C_v is divided into four parts corresponding to the four velocities in the modified Kieffer model:

$$C_v = C_1 + C_2 + C_3 + C_4, \quad (\text{C.1})$$

where suffix 1 corresponds to longitudinal mode, and the others to the three transverse modes in the order of C_{44} , C_S , and C_{111} . Each partial heat capacity C_n ($n=1, 2, 3, 4$) is given by:

$$C_n = w_n N_A k \{ S(x_i) + K(x_i, x_u) \}, \quad (C.2)$$

where N_A is Avogadro number, k Boltzmann constant, and w_n mode weight ($w_1=1$; $w_2=12/13$; $w_3=6/13$; $w_4=8/13$). The $S(x_i)$ and $K(x_i, x_u)$ are heat capacity function to acoustic and optic branches, respectively:

$$S(x_i) = 3 \left(\frac{2}{\pi} \right)^3 \int_0^{x_i} \frac{[\sin^{-1}(x/x_i)]^2 x^2 e^x}{(x_i^2 - x^2)^{1/2} (e^x - 1)^2} dx; \quad (C.3)$$

$$K(x_i, x_u) = \int_{x_i}^{x_u} \frac{x^2 e^x}{(x_u - x_i)(e^x - 1)^2} dx. \quad (C.4)$$

The tables of $S(x_i)$ and $K(x_i, x_u)$ are given in Kieffer's (1979 c) paper, although the definition of $S(x_i)$ in her paper (Eq.22) should be corrected by multiplying by a factor of 3. Non-dimensional frequencies x_i , x_l , x_u are defined in the modified Kieffer model for each velocity v_n :

$$x_i = (2\hbar/\pi k)(6\pi^2 N_A/V)^{1/3} v_n \quad (C.5)$$

$$x_l = (M_h/M_l)^{1/2} x_i \quad (C.6)$$

$$x_u = (1 + M_h/M_l)^{1/2} x_i, \quad (C.7)$$

where V is molar volume, M_h and M_l atomic weight of heavier and lighter ions, respectively.

The thermal energy $U(T, V)$ (T , absolute temperature) is described as summation of partial thermal energy U_n :

$$U(T, V) = \sum_n U_n(T, V) = \sum_n w_n N_A k T \{ A(x_i) + O(x_i, x_u) \}, \quad (C.8)$$

where $A(x_i)$ and $O(x_i, x_u)$ are acoustic and optic thermal energies, respectively:

$$A(x_i) = \left(\frac{2}{\pi} \right)^3 \int_0^{x_i} \frac{[\sin^{-1}(x/x_i)]^2 x}{(x_i^2 - x^2)^{1/2} (e^x - 1)} dx; \quad (C.9)$$

$$O(x_i, x_u) = \int_{x_i}^{x_u} \frac{x}{(x_u - x_i)(e^x - 1)} dx. \quad (C.10)$$

The pressure is given by a generalized Mie-Grüneisen equation of state:

$$P = P_0 + P_T = P_0 + \sum_n \gamma_n U_n/V, \quad (C.11)$$

where P_0 is pressure at 0 K, and γ_n is mode Grüneisen parameter defined as

$$\begin{aligned} \gamma_n &= -[\partial \log(x_{in})/\partial \log(V)]_T \\ &= 1/3 - [\partial \log(v_n)/\partial \log(V)]_T = (K_T C'_n / 2C_n) - 1/6, \end{aligned} \quad (C.12)$$

where the C_n in Eq. (C.12) is not heat capacity but an elastic constant corresponding to v_n . The Grüneisen constant is given as

$$\begin{aligned}\gamma &= -[\partial \log(T)/\partial \log(V)]_S \\ &= V(\partial P/\partial T)_V/C_v = \sum_n \gamma_n C_n/C_v.\end{aligned}\quad (\text{C.13})$$

Further, the thermal expansivity β is described as

$$\beta = \gamma C_v/VK_T = \gamma C_v/(VK_S - \gamma^2 C_v T). \quad (\text{C.14})$$